HireStream-HP — Phase II: Proposed Scope

Discussion note · 16 July 2026
Build v0.9.47

POSITION TODAY

The portal uses no AI at all

Every feature delivered to date — matching, compliance, readiness scoring, the job importer — is deterministic, rule-based code. There are no AI or LLM calls anywhere in the running system. This is a deliberate position, not a gap: the rules are auditable, explainable to a regulator, and behave identically every run. Part A below proposes AI only where rules provably cannot reach. Part B covers the two modules HPSEDC carried out of UAT-03.

Voice input: what exists today

The blue-collar form and the grievance form already accept **voice input**, and it is Hindi-aware. It uses the **browser's built-in Web Speech API** — `window.SpeechRecognition` — not AI, and not a component we built. It is honest to demonstrate this today. Two limits should be understood before it reaches production:

Limitation 1 — coverage: not available on iPhone

The mic only renders where the browser supports speech recognition (Chrome, Edge). **iOS Safari does not.** An iPhone candidate sees no mic at all. For a blue-collar audience this is a coverage gap, not a cosmetic one.

Limitation 2 — data residency: audio leaves the boundary

Chrome's implementation of the Web Speech API **streams the captured audio to Google's servers** for transcription. On an HPSEDC portal, under DPDP, that is a third-party data flow no candidate has consented to. Acceptable for demonstration; it should be closed before production. On-premise speech recognition (item 1 overleaf) removes both limits at once.

DEPLOYMENT MODEL

On-premise, inside the HPSEDC boundary

The candidate data here — Aadhaar, passport, medical fitness, police clearance — must not be sent to a public AI service. Every item proposed below is achievable with a **self-hosted, OpenAI-compatible model endpoint**, keeping all personal data inside the boundary. Model quality matters less than custody here.

A self-hosted endpoint is already configured in the codebase

`server/services/system-config.service.ts` already carries an `llm_triage` block pointing at a self-hosted **Nexus** endpoint (`nexus.osipl.dev/v1`), running `mistral-7b-instruct` , marked "no auth (self-hosted)". It is **configuration only** — the feature is disabled and no code consumes it yet — but the integration shape is already the right one, and Phase II would light it up rather than introduce anything new. *To confirm with HPSEDC: whether this is the same on-premise platform intended for Phase II.*

Phase II — proposed scope

PART A — WHERE AI ADDS VALUE

Each area addresses a limit we have actually hit in the delivered system — not a place AI could theoretically be applied.

#	Area	What it solves	Why rules cannot
1	**Voice-first data entry** On-prem speech recognition, Hindi / Pahari	A candidate speaks their details instead of typing. Works on **every** device including iPhone, and the audio never leaves the HPSEDC boundary.	The browser API is the constraint: we cannot change its device coverage or where it sends audio.
2	**Duplicate detection by meaning** Education, certifications, skills	Recognises "First Aid Training" and "Basic First Aid Workshop" as the same thing and stops the second entry.	Text normalisation now catches spelling and format ("10th" vs "10th Grade"). It cannot catch meaning. More rules will not close this.
3	**Job importer auto-mapping** Government Excel / CSV intake	Reads whatever column layout arrives, maps it to portal fields and normalises job titles and countries — with no human mapping each file.	Every department sends a different layout. Rules must be written per format, in advance.
4	**Ask HPSEDC — triage & assisted replies** Support desk	Answers repeat questions instantly, drafts replies for staff to approve, routes genuine grievances to a human.	Candidates ask the same question in a hundred phrasings, in two languages.
5	**Document understanding** Passport · Aadhaar · PCC · medical	Reads uploaded documents, extracts expiry dates and pre-fills the compliance record feeding deployment readiness.	Readiness is only as good as hand-typed dates. Today a stale certificate can sit unnoticed behind a "verified" status.

If only two are funded: 1 and 4. Item 1 changes *who can use the portal*; item 4 is closest to shippable — the message thread, unread alerts and staff inbox landed in this build. Both keep data on-premise.

PART B — CARRIED FORWARD FROM UAT-03 · CONVENTIONAL MODULES, NOT AI

#	Module	Phase II intent	Status at UAT-03 · what it needs
20	**WhatsApp notifications** "Integrate WhatsApp notifications and communication functionality"	Status updates, interview reminders and document requests on the channel blue-collar candidates actually read. Today: in-app and email only.	**Deferred to Phase 2** — as recorded. Needs Meta WhatsApp-Business verification of the HPSEDC entity, a messaging partner (BSP) and template pre-approval. Externally gated, not a build constraint.
18	**Fee / payment module** "Include a Fee section with relevant details"	To be defined with HPSEDC. Any fee charged must be transparent, receipted and auditable — a published schedule is itself an anti-fraud control.	**Closed as out of scope** — "no candidate-side fee in this government programme". Reopening reverses that position, so the policy question comes before the build.

Item 18 — please settle the policy before the scope

HPSEDC's position at UAT-03 was that this programme charges candidates **nothing**, and the portal matches: no fee exists in it, and the anti-fraud advisory tells candidates anyone demanding money is acting improperly. A payment module **reverses that** and weakens the advisory unless the two are reconciled. Confirm *who pays whom* — candidate, employer or agency — before any estimate.